

# Distance scaling of the heavy-hex subsystem code on IBM Fez: a pre-registered test of a calibration-based noise model

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(Dated: October 9, 2026)

We report a pre-registered quantum-memory experiment with the heavy-hex subsystem code at distances 3 and 5 on IBM’s 156-qubit Heron r2 processor `ibm_fez`, and use it to test whether a circuit-level noise model built from IBM’s published calibration predicts the hardware. The circuits, qubit placement, analysis and decision rules were fixed in code before the first job, and a prediction for every job was committed before the job was submitted. Over five days and ten repetitions on the only two distance-5 placements available each day, the distance-5 patch fails more often than the distance-3 patch in X memory at 59 of 60 combinations of repetition and round count (the sixtieth differs by one shot in 2,500), and the error-suppression factor is  $\Lambda = 0.55 \pm 0.03$ , against 0.51 predicted. The model reproduces  $\Lambda$  once the day-to-day scatter of the calibration is allowed for, but underestimates the logical error rates behind it, by about a factor of two in Z memory, and fails the pre-registered agreement test. Flag and relay qubits, which should read zero, fire in persistent streaks lasting about  $30\mu\text{s}$ , more often on qubits that take part in more two-qubit gates and in correlated pairs, as expected for leakage out of the computational subspace that IBM’s measure-and-flip reset does not remove. A leakage-aware Pauli-frame simulation fitted only to detection statistics closes a third to a half of the Z-memory gap. Predicting logical error rates on this hardware from its calibration requires leakage, reset errors and the calibration’s own drift.

## I. INTRODUCTION

Quantum error correction pays off only if enlarging a code lowers its logical error rate. The standard figure of merit is the error-suppression factor

$$\Lambda = \varepsilon_3/\varepsilon_5, \quad (1)$$

the ratio of the logical error per round at code distances 3 and 5;  $\Lambda > 1$  means the larger code is better [1, 2]. Google’s surface code reached  $\Lambda \approx 1.04$  on Sycamore [1] and  $\Lambda = 2.14 \pm 0.02$  on Willow [2], and on IBM’s square-lattice Nighthawk processor a distance-5 surface code recently beat distance 3 in Z memory, 0.98% against 1.26% per round, after defective components were excluded and leaked shots post-selected away [3].

IBM’s Heron processors instead use a heavy-hex lattice, in which no qubit has more than three neighbours. The heavy-hex subsystem code was designed for this connectivity [4], using flag qubits [5] to measure its weight-four gauge operators. It has been run at distance 3 on a 27-qubit Falcon processor [6] and on Heron r2 [7], and at distance 5 on Heron r3 in an appendix of a study of the dynamic compass code [8]. Other codes embedded in heavy-hex hardware have generally performed worse at larger distance [9, 10]; a SWAP-embedded surface code reached  $\Lambda \approx 1.2$ – $1.5$  in the basis protected by its enlarged distance (1.0–1.1 against the best distance-3 subpatch) but 0.67–0.77 averaged over bases [11]. A comparative study found SWAP-embedded surface codes more promising than the native heavy-hex code for this lattice [12].

On such hardware the sign of  $\Lambda$  is rarely in doubt: simulations from the device calibration put it below one before any data are taken (Sec. VI A). The more useful

question is whether those simulations get the numbers right. Calibration-based models have missed on IBM hardware before: a Stim model underestimated errors on Nighthawk [3], one overestimated distance-3 errors in Ref. [9], reproducing Heron data required fitted reset and readout errors [7], and on `ibm_sherbrooke` a single error rate fitted to detector statistics predicted logical errors better than per-qubit benchmarks [13].

Here we test such a model prospectively. We ran the heavy-hex code at distances 3 and 5 on `ibm_fez` for five days, placing the patches each day from IBM’s latest calibration and committing, before each job, the circuits, the calibration and a prediction made from them. The only model input the calibration does not supply, the dephasing left by our dynamical-decoupling pulses, is measured by idle circuits in the same job. We find  $\Lambda = 0.55 \pm 0.03$  in X memory, close to the predicted 0.51; but the logical error rates themselves are underestimated, most strongly in Z memory, and the pre-registered agreement test fails. An exploratory analysis of the flag and relay readouts identifies leakage as the largest single cause of the miss.

## II. CODE AND CIRCUITS

The  $[[d^2, 1, d]]$  heavy-hex code is a hybrid: it corrects Z errors as a surface code and X errors as a Bacon–Shor code [14] (we follow the labels of Ref. [6]; Ref. [4] exchanges X and Z). The surface-code half measures plaquettes of fixed weight, so below a threshold a larger patch corrects more. The Bacon–Shor half measures stabilizers that are products of gauges along entire column pairs, of weight  $2d$ ; with the same gauges measured every round it has no threshold [4, 15], because a larger patch’s stabilizers accumulate more errors per round. X memory

fails only through Z errors, so it is the test of scaling. Z memory fails through X errors and serves mainly as a test of the noise model. Table I lists the resources at each distance.

A round measures every gauge of the memory basis and then every gauge of the other basis, and lasts  $7.9\mu\text{s}$  at both distances because gauge measurements are interleaved rather than queued. Before the first round, the gauges of the other basis are measured once to give its stabilizers a reference. Each X gauge is measured with one ancilla and two CNOTs. A bulk Z gauge acts on four data qubits that its ancilla cannot reach on heavy-hex; it borrows the two neighbouring X ancillas, which carry the parities of their data pairs to the Z ancilla one qubit at a time and are then uncomputed. Following Refs. [4, 6] we call these borrowed qubits flags, although here they act as bridges: no fault spreads through them onto the data, they read 0 unless an X fault hit them, and in Z memory they serve as extra detectors. The gate order of the 12 CNOTs in each bulk Z gauge ensures that a single fault reaches at most one data qubit, or two in the same column, neither of which reduces the distance. Stim [16] finds circuit distance 3 and 5 in both bases for the circuits that were run, so minimum-weight matching reaches full distance without flag-conditioned decoding. (An earlier version of these circuits, with flags left in random states, reached only 2 and 3 in X memory.) Edge Z gauges reach their data through relay qubits [4, 6].

The circuits are compiled to Fez’s native gates (CZ,  $\sqrt{X}$ ,  $X$  and virtual  $R_Z$ ), with each CNOT realized as  $H\cdot CZ\cdot H$  on its target and every idle period written as an explicit delay. Gates start as soon as their qubits are free, except resets, which are scheduled immediately before the qubit is next used. (Qiskit’s default as-late-as-possible scheduling would have left data qubits in a rotated basis during readout.) Data qubits spend  $6.7\mu\text{s}$  of each round idle while ancillas are read out ( $1.66\mu\text{s}$ ) and reset ( $1.68\mu\text{s}$ ), twice per round. Every idle period of 500 ns or more receives two  $X$  pulses at one and three quarters of its length [17, 18]. IBM’s own decoupling and Pauli twirling are disabled, so that the device executes exactly the circuit we simulate.

TABLE I. The two patches.

|                                        | $d = 3$ | $d = 5$ |
|----------------------------------------|---------|---------|
| Data qubits                            | 9       | 25      |
| Qubits used (with ancillas and relays) | 23      | 65      |
| X stabilizers, weight 4 + weight 2     | 2 + 2   | 8 + 4   |
| Z stabilizers, weight $2d$             | 2       | 4       |
| Readouts per round                     | 18      | 56      |

### III. PLACEMENT AND RUN DESIGN

A distance-3 patch fits on Fez in 30 places and a distance-5 patch in 12, each in two orientations. Each day, every place that uses a part IBM marks faulty, reports at  $\geq 50\%$  error, or reports without  $T_1$  or  $T_2$  is discarded; working but weak parts are kept and listed. Places are scored by the summed CZ, readout and reset errors of one round and the  $T_1$  decay of the data qubits. Each day’s calibration defines up to two repetitions (reps), one per clean distance-5 place, best first. Each rep’s distance-3 patch takes the best clean place not yet used that day: beside the distance-5 patch in the same job where one fits, otherwise in a separate job submitted immediately after.

On every run day exactly two distance-5 places were clean, and they overlap on 51 of their 65 qubits (Fig. 1). Rep 1 therefore ran both patches in one job; for rep 2, no clean distance-3 place fitted beside the distance-5 patch, so its distance-3 patch ran separately on the best clean place on the chip. Each job contains X- and Z-memory circuits at  $n = 1, 2, 3, 4, 6$  and 8 rounds (2,500 shots each) and ten idle-test circuits (Sec. IV; 1,250 shots each). Fifteen jobs ran over five days, each charged 15 s of QPU time, for a total of 225 s against a pre-set budget of 300 s. IBM did not recalibrate during any job.

### IV. NOISE MODEL AND PRE-REGISTERED ANALYSIS

#### A. Prediction

Each prediction is a Stim simulation of the exact compiled circuits, gate by gate and delay by delay. Each CZ receives two-qubit depolarizing noise at  $5/4$  of IBM’s reported error for its coupler, and each  $\sqrt{X}$  and  $X$  one-qubit depolarizing noise at  $3/2$  of the qubit’s reported error, converting average gate infidelity to a depolarizing probability. Each measurement and reset flips with the qubit’s reported readout error. A delay of length  $t$  is the Pauli-twirled amplitude- and phase-damping channel [19],

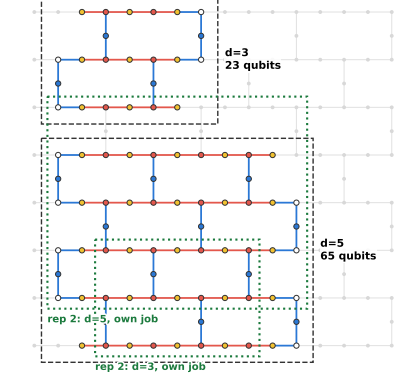
$$\begin{aligned} p_x &= p_y = \frac{1}{4}(1 - e^{-t/T_1}), \\ p_z &= \frac{1}{2}(1 - e^{-t/T_2}) - \frac{1}{4}(1 - e^{-t/T_1}), \end{aligned} \quad (2)$$

with  $T_2 \leq 2T_1$ . In a decoupled delay  $T_2$  is replaced by

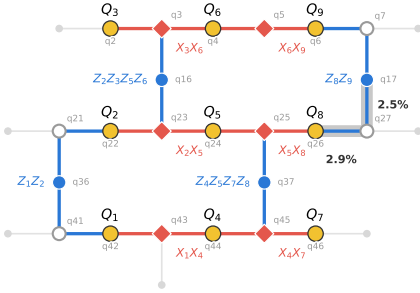
$$\frac{1}{T_2^{\text{DD}}} = \frac{1}{2T_1} + f \left( \frac{1}{T_2} - \frac{1}{2T_1} \right), \quad (3)$$

where  $f$  is the pure-dephasing rate left by the pulses in units of the rate implied by IBM’s Hahn-echo  $T_2$ :  $f = 0$  leaves only  $T_1$  decay and  $f = 1$  recovers echo dephasing.  $f$  is the model’s one input not supplied by the calibration. Each job measures it with idle tests: X-memory circuits in which every CZ is replaced by a delay of equal length,

(a) the day's reps on the chip: rep 1 drawn, the rest boxed



(b) rep 1, d = 3: 9 data qubits, 23 in all



(c) rep 1, d = 5: 25 data qubits, 65 in all

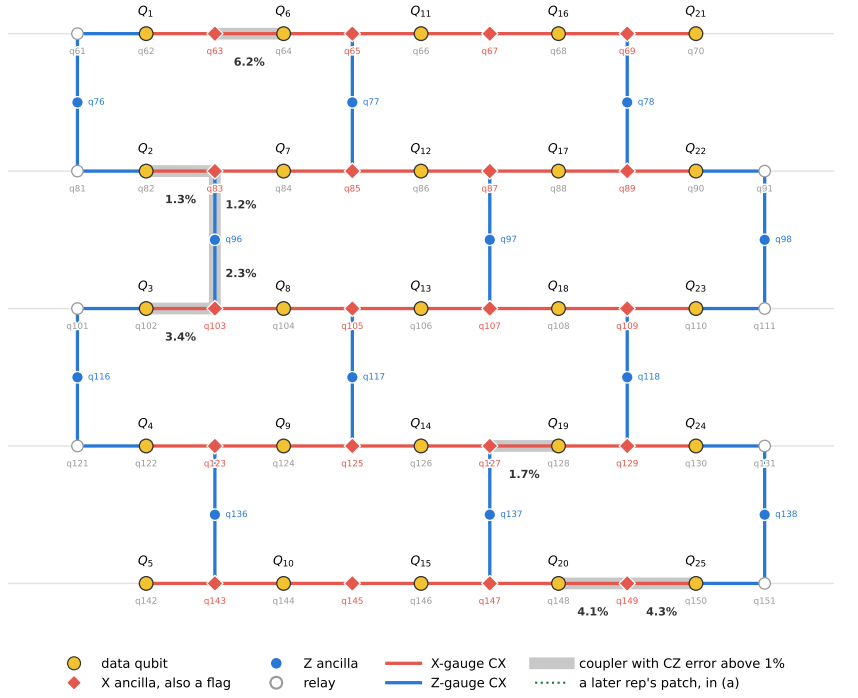


FIG. 1. The two reps planned on IBM's calibration of 3 October 2026, 20:05 UTC (day 1). (a) Placement on the chip: rep 1 drawn, rep 2's patches boxed. (b) Rep 1's distance-3 patch and (c) its distance-5 patch;  $Q$  labels are code qubits and  $q$  numbers Fez qubits. Shaded couplers have CZ error above 1%. The same two distance-5 places were the only clean ones on every run day.

read out in X and in Y, so that a frequency offset, which rotates X into Y, cannot mimic dephasing (Appendix C). Leakage, crosstalk, coherent errors and drift are not modelled. Chip and simulated shots are decoded identically, by minimum-weight perfect matching [20] on the detector error model Stim derives from the same circuit at the measured  $f$ .

## B. Decision rules

For each patch and basis, the logical error  $P(n)$  is fitted as  $1 - 2P(n) = A(1 - 2\varepsilon)^n$  by weighted least squares on  $\ln[1 - 2P(n)]$ , omitting points with  $P(n) \geq 0.5$ ;  $A$  absorbs preparation, readout and time-boundary effects. Each rep's  $\Lambda$  in X memory is called with a one-sided test at nominal 5% in each direction ( $\Lambda \mp 1.645\sigma$  compared with 1). The run's answer is  $\Lambda$  pooled over reps with inverse-variance weights, its standard error widened by  $\sqrt{\chi^2/\nu}$  when the reps scatter more than their errors allow and tested with Student's  $t$  for  $\nu = R - 1$  degrees of freedom. The model agrees with a rep in X memory if  $\Lambda$ ,  $\varepsilon_3$  and  $\varepsilon_5$  each lie within twice the combined observed and predicted standard error of their predictions; over the run, the test is a  $\chi^2$  sum over reps against its 95th

percentile. Z memory is reported but not tested. Side checks for drift within a job, for a rising detection rate over rounds (leakage or heating), and for individual detectors firing far above prediction are reported without affecting the decisions. Appendix B gives the details and the simulated error rates of these rules.

## C. Record

The design, code and decision rules were fixed in code before the first job: the multi-day design and the pooling rule were committed 31 and 13 minutes before it, and the code was released as v0.2.0 and archived by Zenodo [21] four minutes before it, the run's one third-party timestamp. Each day's folder of calibration, compiled circuits and prediction was committed and pushed before that day's first job, and the submission software refuses a job whose folder is not. The analysis code did not change during the run: all fifteen jobs record the same version. Every planned job was run and none was repeated or excluded. Appendix A gives the timeline.

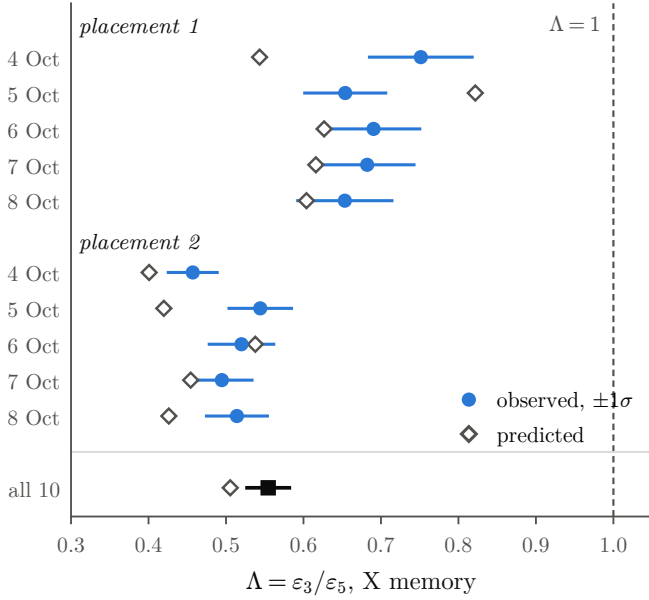


FIG. 2.  $\Lambda$  in X memory for each rep, observed (circles,  $\pm 1\sigma$ ) and predicted at the job’s measured  $f$  (diamonds), grouped by placement, and the pooled value with its widened error.

## V. RESULTS

All ten reps call distance 5 worse in X memory, with  $\Lambda$  between 0.46 and 0.75 (Fig. 2). Pooled over the reps,  $\Lambda = 0.55 \pm 0.03$ , against 0.51 predicted. The reps scatter more than their errors ( $\chi^2/\nu = 3.9$ ); with the error widened accordingly and  $t_9 = 1.83$ , the pooled rule calls distance 5 worse. The result does not depend on the fitting procedure: in X memory the distance-5 patch has the higher logical error in 59 of the 60 combinations of rep and round count, and in the sixtieth (7 October, rep 1, one round) the two differ by a single shot in 2,500 (Fig. 3).

The ten reps are essentially two placements measured on five days each: the distance-5 patches occupied the same two places every day, and rep 1’s distance-3 patch one place, while rep 2’s distance-3 patch occupied three different places over the run, all within rep 1’s distance-5 footprint. In X memory,  $\Lambda$  is  $0.68 \pm 0.03$  on placement 1 and  $0.50 \pm 0.02$  on placement 2 (predicted 0.64 and 0.45 on average), and within each placement the five days agree ( $\chi^2/\nu = 0.4$  and  $0.8$ ; Table II). The device’s X-memory  $\Lambda$  is therefore stable from day to day, and the excess scatter between reps reflects placement; Z memory varies more ( $\chi^2/\nu = 1.9$  and  $2.7$  within placements). Because the pooled rule treats reps as independent, its error bar is too narrow for a statement about the device as a whole: taking the placement as the unit ( $n = 2$ ) gives  $\Lambda = 0.59 \pm 0.09$ , which the same test leaves undecided (one-sided  $p \approx 0.07$ ). The conclusion that distance 5 is worse is firm for these two placements, which were the only clean ones on every run day, and suggestive for Fez in general. Other reasonable estimators give 0.55–0.60 (Appendix B).

## VI. MODEL VERSUS DEVICE

The model predicts  $\Lambda$  reasonably well but not the error rates that form it.  $\varepsilon_3$  in X memory exceeds its prediction in every rep, by 1–56%, and the run-level tests fail for  $\Lambda$  ( $\chi^2 = 25.0$  against a 95% bound of 18.3 for ten reps),  $\varepsilon_3$  (52.1) and  $\varepsilon_5$  (19.8); four of ten reps agree on all three quantities and eight on  $\Lambda$  alone. Z memory errs about twice as often as predicted at distance 3 ( $\chi^2 = 1240$  for  $\varepsilon_3$  and 271 for  $\varepsilon_5$ ), and its  $\Lambda$  fails as well (35.2). Because Z memory does not depend on  $f$ , its miss points at the calibrated CZ, readout and  $T_1$  inputs or at physics outside the model.

Part of the X-memory disagreement reflects the calibration rather than the chip. On placement 1 the predicted  $\Lambda$  moved between 0.54 and 0.82 over the five days while the measured value stayed between 0.65 and 0.75. The pre-registered bound allows for shot noise, the uncertainty in  $f$  and recalibration during a job, but not for this day-to-day scatter of IBM’s numbers. As a post hoc check, we add to each predicted quantity its relative standard deviation over the five days of predictions on placement 1 (16% for  $\Lambda$ , 9% for  $\varepsilon_3$ , 8% for  $\varepsilon_5$ ).  $\Lambda$  then passes ( $\chi^2 = 7.8$  against 18.3), robustly so: any relative scatter above about 6% suffices.  $\varepsilon_5$  passes marginally (13.4), and would fail with placement 2’s smaller prediction scatter;  $\varepsilon_3$  (35.3) and all Z-memory tests still fail. This treats the calibration’s daily changes as noise in the prediction rather than as changes of the device, an interpretation suggested, but not proven, by the device’s own day-to-day stability.

The decoupling pulses removed little dephasing. The idle tests give  $f$  between 0.72 and 1.52 across the fifteen jobs (median 0.99), with a spread across qubits far exceeding the per-qubit uncertainties ( $\chi^2/\nu$  of 12–80 about the weighted mean in every job). Values of  $f$  read back from the memory runs agree with the idle test in ten of fifteen jobs and are higher in the other five, as expected if the idle test, which omits the CZ pulses and keeps neighbouring ancillas in  $|0\rangle$ , underestimates the dephasing during memory. The detection rate rises over the rounds beyond the prediction in 25 of the 40 eight-round runs, and is flagged in every job; drift within jobs is absent (largest change between the first and second halves of a job’s shots:  $2.5\sigma$ ).

### A. Predicted origin of $\Lambda < 1$

Before the run, the model already placed  $\Lambda$  below one for two reasons. First, placement: the distance-5 patch must take whichever of its 12 places is clean, and on day 1 its best place contained a coupler with 6.2% CZ error, a qubit with 14% readout error and two qubits with  $T_2$  of 5–6  $\mu\text{s}$ . Moving the distance-3 patch onto the distance-5 patch’s qubits raises the simulated  $\Lambda$  from 0.54 to 0.65–0.70, whereas setting only the weak parts of both patches to the chip median gives 0.61–0.62. Second, the

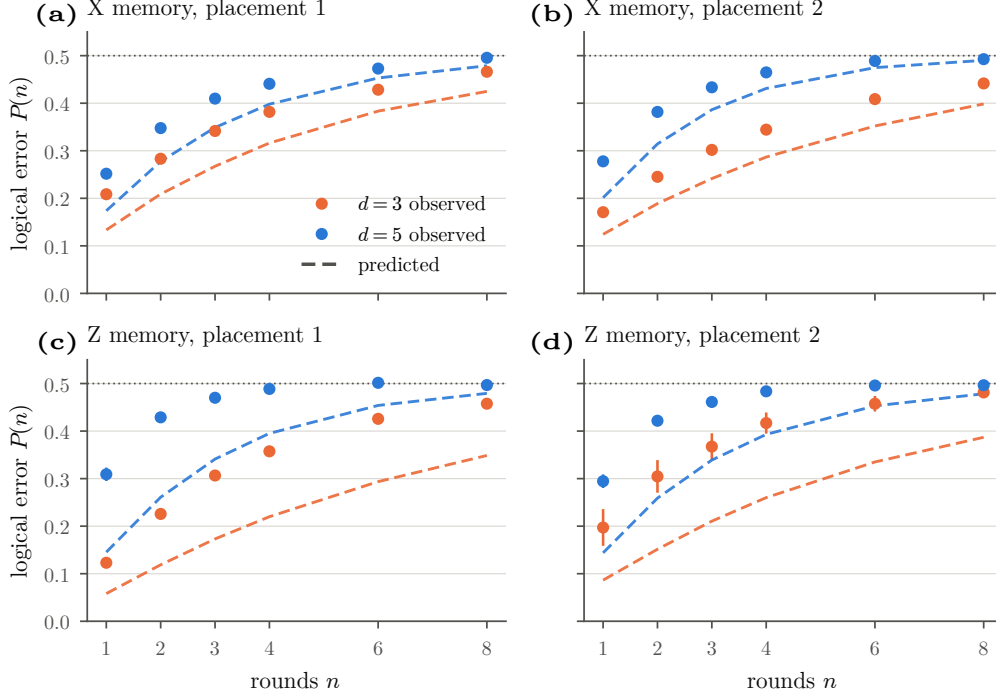


FIG. 3. Logical error  $P(n)$  after  $n$  rounds on each placement, mean over the five days ( $\pm$  standard error over days), with the prediction at each job's measured  $f$  averaged likewise (dashed). The chip exceeds the prediction from the first round; in Z memory the distance-3 error is about twice the prediction. Distance-5 X memory approaches 50% by six rounds.

TABLE II. Logical error per round and  $\Lambda$  by placement, pooled over the five days ( $\pm$  one standard error, widened as in Sec. IV), with the mean prediction at the measured  $f$  and its range over the days.

|                |       | $\varepsilon_3$ (%) | $\varepsilon_5$ (%) | $\Lambda$        |
|----------------|-------|---------------------|---------------------|------------------|
| X, placement 1 | obs.  | $12.5 \pm 0.3$      | $18.4 \pm 0.5$      | $0.68 \pm 0.03$  |
|                | pred. | 10.2 (9.6–11.8)     | 16.1 (14.3–17.9)    | 0.64 (0.54–0.82) |
| X, placement 2 | obs.  | $11.0 \pm 0.4$      | $21.9 \pm 0.6$      | $0.50 \pm 0.02$  |
|                | pred. | 8.6 (7.7–10.2)      | 19.2 (18.3–20.1)    | 0.45 (0.40–0.54) |
| Z, placement 1 | obs.  | $13.7 \pm 0.3$      | $29.8 \pm 1.7$      | $0.44 \pm 0.03$  |
|                | pred. | 7.1 (6.2–8.2)       | 16.6 (15.3–17.4)    | 0.43 (0.35–0.49) |
| Z, placement 2 | obs.  | $15.6 \pm 1.3$      | $26.3 \pm 1.0$      | $0.63 \pm 0.05$  |
|                | pred. | 8.4 (7.7–8.8)       | 16.5 (15.4–18.3)    | 0.51 (0.42–0.57) |

error rates: with every CZ, readout,  $T_1$  and  $T_2$  at the chip median, the simulated  $\Lambda$  is 0.94 at  $f = 0$  and 0.86 at  $f = 1$ ; halving the CZ error raises it to 1.10 at  $f = 0$ . Figure 4 shows the same scan on a uniform chip at the medians of 8 October: X memory reaches  $\Lambda = 1$  at 0.75 times today's error rates and Z memory only at 0.31, and the latter crossing would not persist at larger distance because the Bacon–Shor half has no threshold.

## VII. DIAGNOSIS: LEAKAGE

Everything in this section is exploratory: it was decided after the run, uses only the run's shots and IBM's

published calibrations, and changes nothing above.

*a. Flags remember.* In Z memory every flag and relay should read 0 in each round, and in the calibration model successive rounds are independent. On the device, a flag or relay that fires in one round fires in the next 14.8 percentage points more often than one that did not, and the excess decays over about  $40 \mu\text{s}$  (Fig. 5a); the model gives zero at every lag. The excess appears in 319 of the 320 readout series of the run, and runs of five or more consecutive firings are ten times as frequent as the model allows (0.33% of shots against 0.035%).

*b. A persistent hidden state.* A two-state hidden Markov model, with a normal and a persistent state, fits 97% of the readout series better than independent rounds (log-likelihood gain above 10, median 327) and finds no

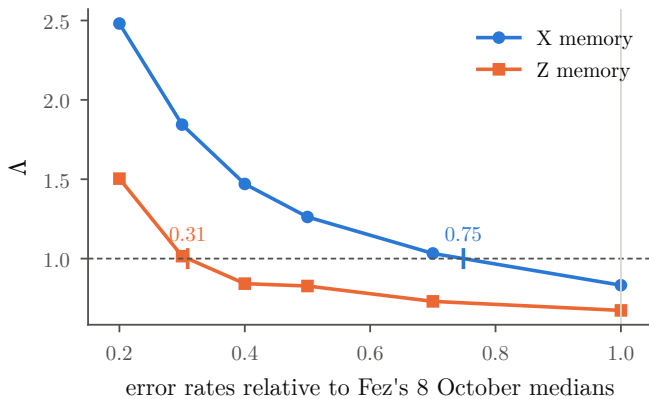


FIG. 4. Simulated  $\Lambda$  on a uniform chip at Fez's 8 October median error rates, with all error rates scaled together ( $f = 1$ ). Ticks mark where  $\Lambda$  crosses 1.

such state in shots sampled from the model. The persistent state is entered in about 2.5% of rounds, lasts about  $31\ \mu\text{s}$  (four rounds), and makes the readout fire about half the time. Outside it, flags fire in 4.1% of rounds against the model's 4.4%.

*c. Consistent with leakage.* In transmon processors, leakage out of  $\{|0\rangle, |1\rangle\}$  arises in two-qubit gates and in measurement, persists for tens of microseconds, and spreads to neighbours through two-qubit gates [22, 23]; a reset that measures and conditionally flips the qubit acts only within  $\{|0\rangle, |1\rangle\}$  and does not return a leaked qubit [3]. These observations come from processors with tunable-frequency qubits and a different CZ gate, so their rates need not carry over to Heron, but the persistent state shows the same qualitative features. Its onset rate is about twice as high on flags, which take part in 8.25 CZs per round, as on relays, which take part in three (Fig. 5b); a linear fit attributes  $\approx 0.31\%$  per CZ and  $\approx 0.94\%$  per round to readout and reset, although with only two roles the CZ count cannot be separated from other differences between flags and relays. The rate barely tracks IBM's reported CZ error for the qubit's couplers (Spearman  $\rho = 0.15$ ) and not at all the number of simultaneous CZs on neighbouring couplers ( $\rho = 0.02$ ), so we see no sign of crosstalk from parallel gates, although the circuit varies little in that respect. IBM's binary readout cannot distinguish  $|2\rangle$ , so the identification rests on the persistence, lifetime, gate dependence and spatial correlations rather than on direct observation.

*d. It spreads.* Two flags that read a common data qubit fire in streaks together far more often than chance ( $6.8 \pm 0.4 \times 10^{-4}$  excess co-streak probability, against  $0.6 \times 10^{-4}$  in the model), as do the two flags of one Z gauge, which share only its ancilla ( $8.0 \pm 0.8 \times 10^{-4}$ , against 0.0); unrelated pairs barely do ( $0.5 \times 10^{-4}$ ). Data qubits and Z ancillas therefore leak as well, and a leaked qubit disturbs, or transfers leakage to, the qubits it shares CZ gates with [22].

*e. Resets and readout.* The model has two smaller faults. It assigns each reset the qubit's readout error (about 1.0% at the median), whereas IBM reports a separate reset error about one sixth as large, and it assigns each readout the average of  $P(1|0)$  and  $P(0|1)$ . Flags sit about 3 points above the model on qubits with reported readout error below 2%, close to it between 2% and 5%, and 3 points below it above 5%. This is not regression to the mean from selecting qubits on noisy numbers: binned by the other days' reported values instead of the day's own, the pattern persists, and the day's own value predicts the flags better (correlation 0.51 against 0.43).

*f. A leakage-aware simulation.* Stim cannot represent a qubit outside  $\{|0\rangle, |1\rangle\}$ , so we replayed every job's memory circuits in a Pauli-frame simulator with a leaked flag per qubit, which reproduces the Stim model exactly when leakage is disabled. Each qubit leaks with a fixed probability per CZ; while leaked, its CZs flip the partner with some probability and transfer the leakage with another; it returns after the hidden-Markov lifetime; and resets and readouts use IBM's reset error and its separate  $P(1|0)$  and  $P(0|1)$ . Fitted to six detection statistics only, and to no logical error, the model takes 0.2% leakage per CZ, flips the partner in a quarter of leaked CZs and transfers leakage in 5%. It then matches the detection rates of the flags and of the X-memory stabilizers to within 6%, but leaves the Z-memory stabilizers 11% short. Decoded as the device's shots were, it moves Z memory a third to a half of the way from the calibration model to the device, and overshoots X memory (Table III); replacing the bit flip on the partner by a random Pauli gives the same result. The mean  $\Lambda$  in X memory moves little, from 0.54 to 0.57, against 0.60 on the device (unweighted means over reps). Leakage thus accounts for a large share of the Z-memory excess, but this simple model does not distribute its effect between the bases as the device does, and part of the Z-memory excess has another source.

## VIII. DISCUSSION

$\Lambda$  for the heavy-hex code at distances 3 and 5 on Fez is below one in both bases, as the calibration model predicted, and is set mainly by placement: in X memory the two placements give 0.68 and 0.50, each stable over five

TABLE III. Logical error per round averaged over the fifteen jobs: measured, predicted by the calibration model, and simulated with the fitted leakage model (10,000 shots per circuit), all decoded identically.

|                   | Device | Calibration model | With leakage |
|-------------------|--------|-------------------|--------------|
| X memory, $d = 3$ | 11.9%  | 9.4%              | 13.2%        |
| X memory, $d = 5$ | 20.2%  | 17.6%             | 23.3%        |
| Z memory, $d = 3$ | 15.8%  | 7.6%              | 10.2%        |
| Z memory, $d = 5$ | 27.3%  | 16.6%             | 21.5%        |

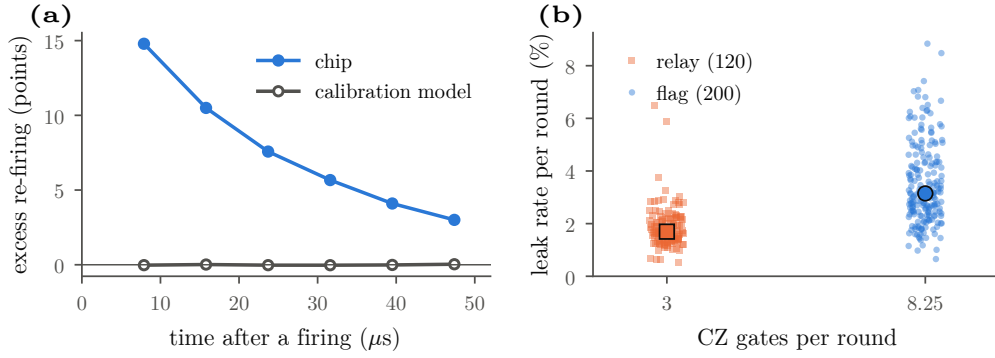


FIG. 5. (a) Excess probability that a flag or relay fires again a given time after it fired, relative to one that did not, averaged over the 320 readout series of eight-round Z memory, for the device and for shots sampled from the calibration model. (b) Per-series rate of entering the persistent state from a two-state hidden Markov fit, against the number of CZ gates per round the qubit takes part in (relays 3, flags 8.25; horizontal jitter added). Large markers are medians.

days. With only two placements, this is firm for them and suggestive for the device.

The calibration model captures the ordering of the placements and, once its own day-to-day scatter is admitted,  $\Lambda$  in X memory. It does not capture the logical error rates. The discrepancies point in one direction: decoupling with two X pulses leaves about echo-level dephasing, the memory runs imply more dephasing than the idle test, and the flags reveal persistent excitations, consistent with leakage, that the resets do not clear and that spread through CZ gates.

Our distance-3 errors are comparable to earlier heavy-hex experiments in X memory and worse in Z memory. Without leakage post-selection, Ref. [6] reports 11.3% per round in X memory and 5.9% in Z memory at distance 3 on a Falcon processor (about 3–4% in Z memory after post-selection), against our 10–14% and 13–24%. On Heron r2, Ref. [7] reduced the error per round from above 10% to 3–4% by measuring the X and Z gauges in parallel and without resets, shortening the round from 11.1 to 3.2  $\mu\text{s}$ ; it attributes the gain mainly to reduced relaxation during measurement and found resets alone to make a negligible difference. Our round spends 6.7  $\mu\text{s}$  of its 7.9  $\mu\text{s}$  on two readouts and two resets, and uses twelve CNOTs per bulk Z gauge; idle decay grows with the first and, in our diagnosis, leakage with both. The natural next steps are a shorter round with parallel gauge measurements, a shorter Z gauge, resets that remove leakage [23], longer decoupling sequences, and twirled gates and readout so that the noise is closer to the Pauli channels a stabilizer model assumes. A calibration-based model of such circuits needs, at a minimum, IBM’s reset errors, readout errors resolved by outcome, the calibration’s day-to-day scatter in its error budget, and a leakage term with its spread.

*a. Limitations.* Only two placements were available, they overlap on 51 of 65 qubits, and they were the same every day. One calibration serves all reps of a day, and a split rep’s two patches run minutes apart. The pre-registered error bars include shot noise, the measured  $f$

and recalibration during a job, but not day-to-day calibration scatter.  $\varepsilon$  and  $\Lambda$  describe one to eight rounds, not asymptotic rates; the distance-5 eight-round points lie near 50% and drop out of the fit in four of ten reps in each basis, and in Z memory on two occasions the six-round point as well, leaving  $\varepsilon_5$  to rounds one to four. The idle test likely underestimates  $f$  during memory. The diagnosis is exploratory, fitted to the run’s own data, with a leakage model whose form is assumed.

## ACKNOWLEDGMENTS

We acknowledge the use of IBM Quantum services for this work. The views expressed are those of the author and do not reflect the official policy or position of IBM or the IBM Quantum team. The circuits, simulations, placement and analysis code were written with the AI assistants Claude (Anthropic) and Cursor, which also helped edit the text; run days 2–5 were prepared and submitted by a script under the same QPU limits, and the exploratory diagnosis was carried out with AI assistance at the author’s direction. The author designed the experiment, checked the code and results, and takes responsibility for the content.

*Data and code availability.* The code, every job’s calibration, compiled circuits, prediction, shots and analysis, the diagnosis scripts and results, and the scripts that draw the figures are available at Ref. [24].

## Appendix A: Pre-registration record

The repository’s history before the run was rewritten twice by force-pushes (24 September and 2 October 2026; the second changed only commit messages); from 3 October onward it matches GitHub’s record of pushes. On 3 October two single-job designs were released (v0.1.0 and v0.1.1) and then replaced; a folder prepared for a single job that day was deleted without being submitted. The

final multi-day design (commit `1c86fce`) and pooling rule (`cd4a68e`) were committed 31 and 13 minutes before the first job, at 21:33 UTC on 3 October, and released as v0.2.0 at 21:29 UTC and archived by Zenodo [21]; the archive matches that commit file for file. Days 2–5 were not released separately. Their record is the git history and GitHub’s log of pushes, which shows each day’s folder pushed 14–15 s before that day’s first submission (and 8 min before on day 1), and the IBM job identifiers, whose embedded timestamps match the recorded submission times. On day 2 the folder was prepared twice, because IBM published a new calibration between preparation and submission; only the calibration’s timestamp changed, and the prediction was identical. Days 2–5 were prepared and submitted by a script added after day 1, which enforces the same per-job and total QPU limits through the unchanged submission code. No copy of the manuscript text was archived before the run: the design and decision rules it describes are those of the code as released. The code under `src/` and its configuration are unchanged from `cd4a68e` to the end of the run, and each job’s recorded code hash is identical.

## Appendix B: Statistical procedures

*a. Fit.*  $\text{Var}(\varepsilon)$  is scaled by the reduced  $\chi^2$  of the fit when it exceeds one and propagated to  $\Lambda$  as  $\text{Var}(\Lambda)/\Lambda^2 = \text{Var}(\varepsilon_3)/\varepsilon_3^2 + \text{Var}(\varepsilon_5)/\varepsilon_5^2$ . At one to eight rounds a single  $\varepsilon$  is an approximation: in simulation of rep 1 the fit has reduced  $\chi^2 = 12.6$  for distance-5 X memory at  $f = 0$ , and  $A \approx 1.04$  in distance-3 Z memory. Omitting the one-round point changes the predicted  $\Lambda$  by 0.01; prediction and data are fitted identically. As an exploratory check,  $\Lambda$  from rounds 2–8 is 0.88, 0.67, 0.62, 0.64 and 0.78 for rep 1 on days 1–5 (predicted 0.53, 0.82, 0.62, 0.61, 0.60) and 0.51, 0.61, 0.55, 0.62 and 0.46 for rep 2 (predicted 0.39, 0.42, 0.54, 0.44, 0.42), each  $\pm 0.1$ .

*b. Power.* In simulation of these fits at 2,500 shots per circuit, a single rep’s  $\Lambda$  has a standard error of about 0.03. A pooled  $\Lambda$  of exactly one is called either way in 2–6% of simulated runs with shot noise alone, 7–20% with twice that scatter and up to 41% with four times; at the predicted  $\Lambda$  every simulated run calls distance 5 worse. If the model is exactly right, the three correlated run-level agreement tests together fail in 6–12% of simulated runs,

and the  $\Lambda$  test detects a mismatch of 0.04–0.06 in every rep with 80% probability.

*c. Agreement bound.* The predicted variance combines simulator shot noise; the uncertainty in  $f$  times the quantity’s slope in  $f$ , from the predictions at  $f = 0$ , the measured  $f$  and  $f = 1$  (or from the line through  $f = 0$  and  $f = 1$  when the measured  $f$  lies within 0.05 of either end or beyond it, in nine of the fifteen jobs); and  $\Delta^2/2$ , where  $\Delta$  is the difference between predictions from the calibrations before and after the job ( $\Delta = 0$  throughout).

*d. Other estimators.* Pooling  $\ln \Lambda$  instead of  $\Lambda$  gives  $0.58 \pm 0.03$ , and the unweighted mean of the reps  $0.60 \pm 0.03$ ; inverse-variance pooling of a ratio is biased low because a smaller  $\Lambda$  has a smaller error.

*e. Detectors.* A detector is flagged if it fires above prediction by more than  $\sqrt{2 \ln N} = 3.81$  standard errors, with  $N = 1428$  detectors per two-patch job; for independent detectors the largest excess has median 3.3 and exceeds this bar in about 9% of jobs. Between 171 and 834 detectors per job exceed it against the calibrations both before and after the job; the largest excesses, up to 5.3 times the predicted rate, are flag detectors in Z memory.

## Appendix C: Measuring $f$

For each data qubit,  $r(n) = \sqrt{\langle X \rangle^2 + \langle Y \rangle^2}$  after  $n$  idle rounds decays through dephasing alone. We fit  $\ln r(n) = a + \ln x_0(n) - f [\ln x_0(n) - \ln x_1(n)]$  by weighted least squares, where  $x_0$  and  $x_1$  are the simulated  $\langle X \rangle$  at  $f = 0$  and 1 and  $a$  absorbs readout error;  $\langle X \rangle^2 + \langle Y \rangle^2$  is corrected for shot noise and points with  $r \leq 0.05$  are dropped. Negative estimates are set to zero, qubits whose  $f$  cannot be determined to  $\pm 0.5$  are excluded, and the median over data qubits is used, with standard error 1.25 times that of the mean. Across all jobs the interquartile range over qubits is 0.46–1.53, and the inverse-variance mean (0.02–0.59) is biased low because larger values of  $f$  are measured less precisely (Spearman  $\rho = 0.76$  between  $f$  and its uncertainty).

## Appendix D: Per-rep results

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TABLE IV. Each rep against the prediction at its measured  $f$  (in parentheses). The last column lists which of  $\varepsilon_3$ ,  $\varepsilon_5$ ,  $\Lambda$  agree under the pre-registered bound.

|   | Day, rep | $\varepsilon_3$ (%) | $\varepsilon_5$ (%) | $\Lambda$              | Agree                       |
|---|----------|---------------------|---------------------|------------------------|-----------------------------|
| X | 4 Oct, 1 | 13.2 (9.7)          | 17.6 (17.9)         | $0.75 \pm 0.07$ (0.54) | $\varepsilon_5$             |
|   | 4 Oct, 2 | 10.1 (8.1)          | 22.0 (20.1)         | $0.46 \pm 0.03$ (0.40) | all                         |
|   | 5 Oct, 1 | 11.9 (11.8)         | 18.2 (14.3)         | $0.65 \pm 0.05$ (0.82) | $\varepsilon_3$             |
|   | 5 Oct, 2 | 12.0 (7.7)          | 22.0 (18.3)         | $0.54 \pm 0.04$ (0.42) | $\varepsilon_5$ , $\Lambda$ |
|   | 6 Oct, 1 | 13.7 (10.4)         | 19.8 (16.6)         | $0.69 \pm 0.06$ (0.63) | $\varepsilon_5$ , $\Lambda$ |
|   | 6 Oct, 2 | 11.1 (10.2)         | 21.4 (19.0)         | $0.52 \pm 0.04$ (0.54) | all                         |
|   | 7 Oct, 1 | 12.1 (9.7)          | 17.8 (15.8)         | $0.68 \pm 0.06$ (0.62) | all                         |
|   | 7 Oct, 2 | 10.5 (8.8)          | 21.2 (19.3)         | $0.49 \pm 0.04$ (0.45) | all                         |
|   | 8 Oct, 1 | 12.5 (9.6)          | 19.1 (15.8)         | $0.65 \pm 0.06$ (0.60) | $\varepsilon_5$ , $\Lambda$ |
|   | 8 Oct, 2 | 11.9 (8.2)          | 23.2 (19.3)         | $0.51 \pm 0.04$ (0.43) | $\varepsilon_5$ , $\Lambda$ |
| Z | 4 Oct, 1 | 12.9 (6.5)          | 26.6 (16.7)         | $0.48 \pm 0.05$ (0.39) | none                        |
|   | 4 Oct, 2 | 13.7 (8.8)          | 23.3 (15.4)         | $0.59 \pm 0.11$ (0.57) | $\varepsilon_5$ , $\Lambda$ |
|   | 5 Oct, 1 | 13.5 (7.0)          | 32.6 (17.0)         | $0.42 \pm 0.04$ (0.41) | $\Lambda$                   |
|   | 5 Oct, 2 | 20.1 (8.5)          | 28.0 (16.1)         | $0.72 \pm 0.11$ (0.53) | $\Lambda$                   |
|   | 6 Oct, 1 | 14.9 (8.2)          | 23.6 (16.6)         | $0.63 \pm 0.10$ (0.49) | $\varepsilon_5$ , $\Lambda$ |
|   | 6 Oct, 2 | 14.9 (8.7)          | 26.6 (16.3)         | $0.56 \pm 0.04$ (0.54) | $\Lambda$                   |
|   | 7 Oct, 1 | 13.6 (7.6)          | 27.9 (15.3)         | $0.49 \pm 0.04$ (0.49) | $\Lambda$                   |
|   | 7 Oct, 2 | 17.0 (7.7)          | 23.2 (18.3)         | $0.73 \pm 0.11$ (0.42) | $\varepsilon_5$             |
|   | 8 Oct, 1 | 13.8 (6.2)          | 34.1 (17.4)         | $0.40 \pm 0.03$ (0.35) | $\Lambda$                   |
|   | 8 Oct, 2 | 23.6 (8.1)          | 26.9 (16.3)         | $0.88 \pm 0.10$ (0.50) | none                        |

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